BIG SYNTAX DOESN'T WANT YOU TO KNOW

MAL is universal? Hardly.

The Menzerath-Altmann law on all 180 UD treebanks

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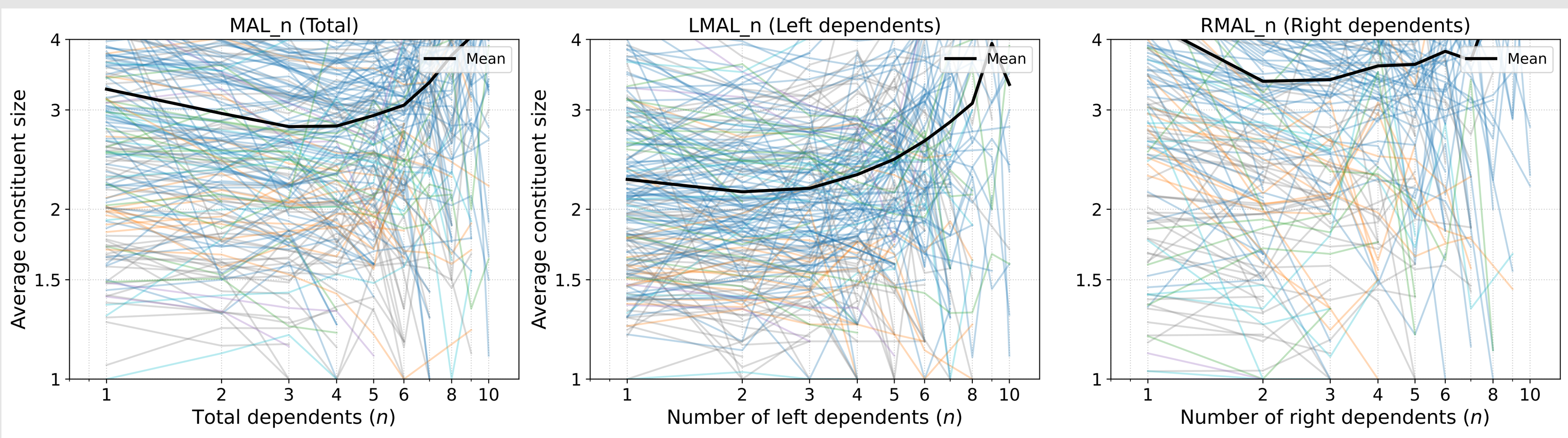
LINGUISTS HATE THIS TRICK

Got more dependents? Just make them smaller!



Wait, it's not a universal?

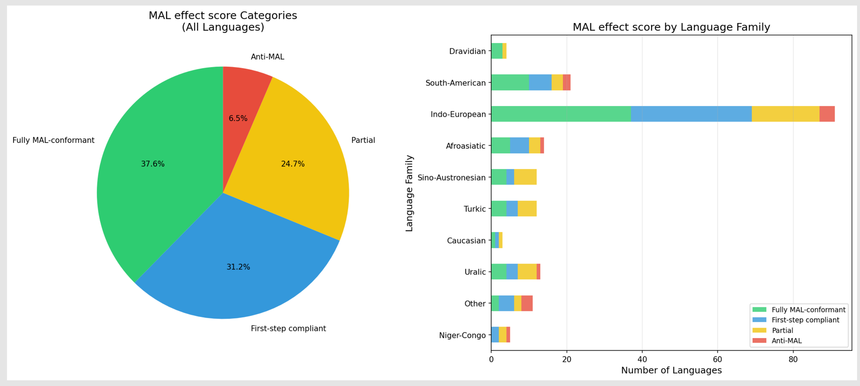
In **most languages, yes.** The average size of a verb's dependents drops as n grows – exactly what Menzerath-Altmann predicts. **Each thin line = one of 186 languages**; the bold line is the cross-linguistic mean.

Left vs right asymmetry

Left dependents stay **flat** as n grows; right dependents **collapse** from 5.2 down to 2.3. **Same law, two different worlds.** The verb is not ambidextrous – it leans right.

WAKE UP, INDO-EUROPEAN!

Some families fail the test!

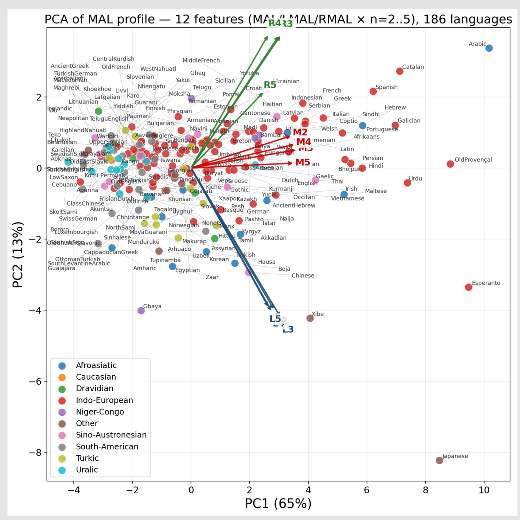


Where do the laws break?

Only 38% of languages are fully MAL-conformant. **6.5% are openly anti-MAL** – and every family hides at least one rebel. Indo-European is no model student.

THE DOTS THEY DON'T WANT YOU TO CONNECT

tiny points, BIG secrets!

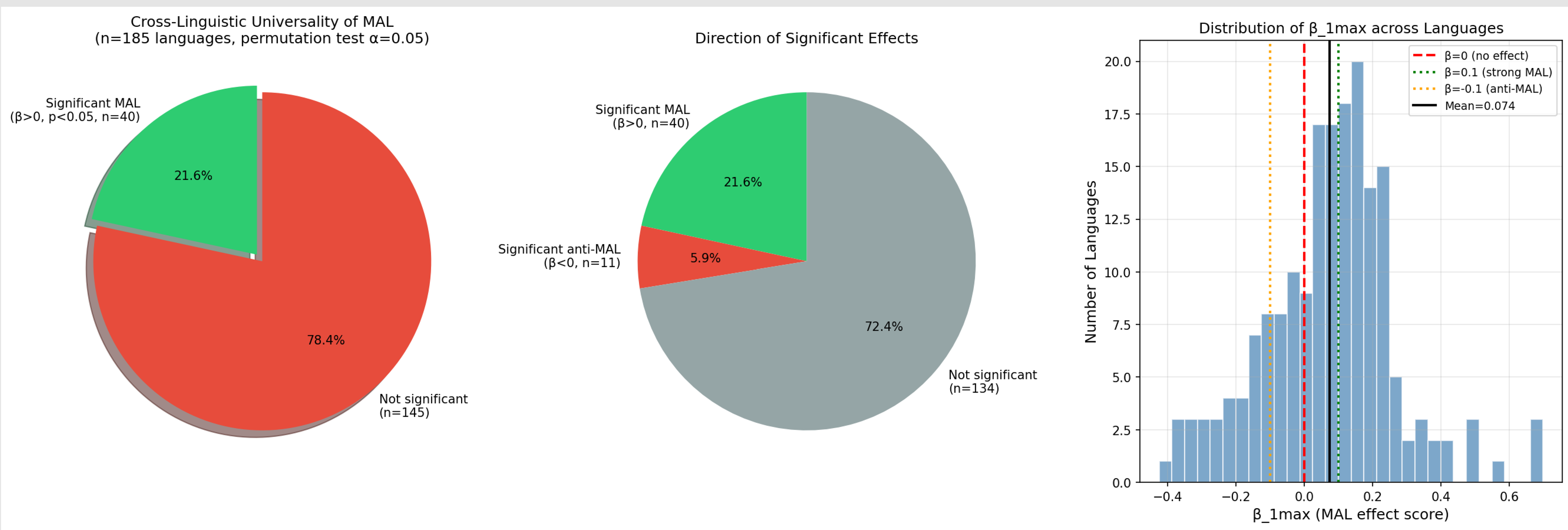


Typological PCA clustering

Every language projected onto a 2D map of its MAL profile (PCA on 12 features: MAL/LMAL/RMAL at $n = 2..5$). **PC1 = 65%, PC2 = 13%.** Indo-European clusters tightly; Japanese, Esperanto and Old Provençal drift out as outliers.

THE 78% NOBODY TALKS ABOUT

Most languages flunk the test.

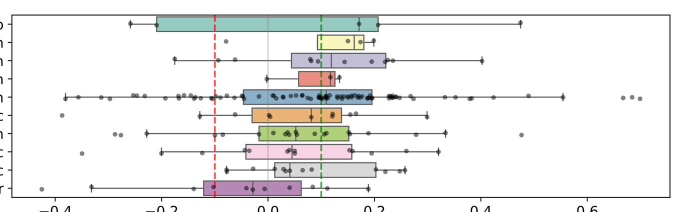


Only 21.6% of UD languages show a statistically significant MAL effect. **5.9%** go the wrong way (anti-MAL). The rest are flat: noise where the textbooks promised a law.

If MAL were a **linguistic universal**, this pie would be one big green slice. It isn't. Welcome to the post-universal era.

FAMILY MATTERS

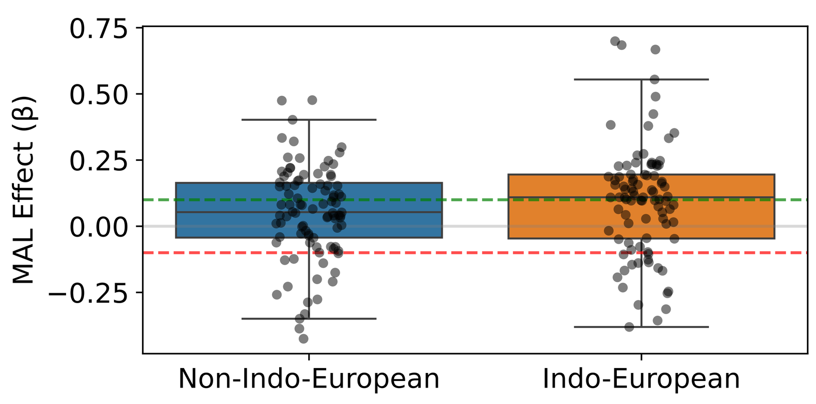
Is MAL just a European bias?



Distribution of the MAL effect (β) across language families. Indo-European clearly dominates the MAL-compliant zone.

THE EURO-CENTRIC LAW

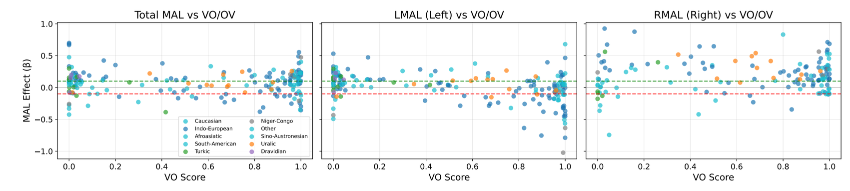
IE vs. The Rest



Indo-European languages exhibit a significantly stronger and more consistent MAL effect compared to all other families combined.

BEFORE AND AFTER

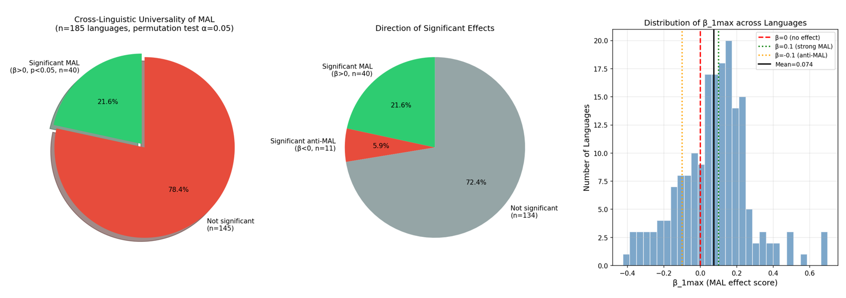
Why left and right differ



RMAL vs VO/OV order. The right-side collapse is strongest when verbs precede objects. The syntax restricts dependent sizes asymmetrically.

UNIVERSAL TRUTH OR STATISTICAL ARTIFACT?

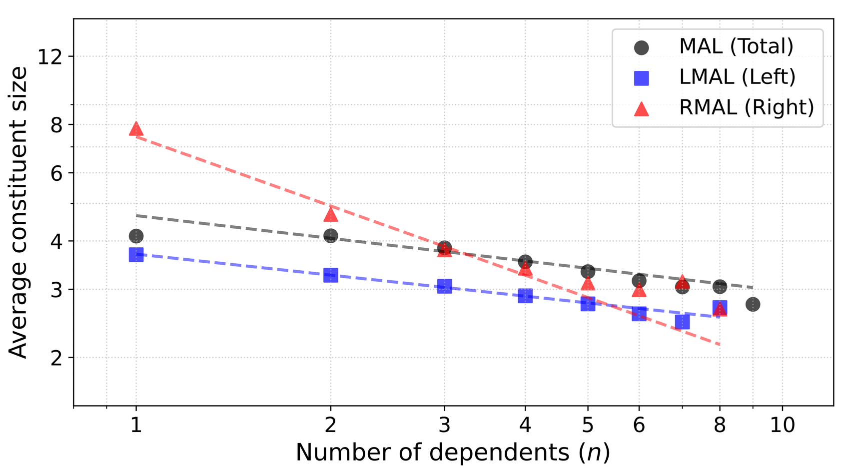
The Beta spectrum



Most β values are indeed negative (MAL-compliant), but a massive chunk straddles zero. Is it a law, or just a heavy statistical lean?

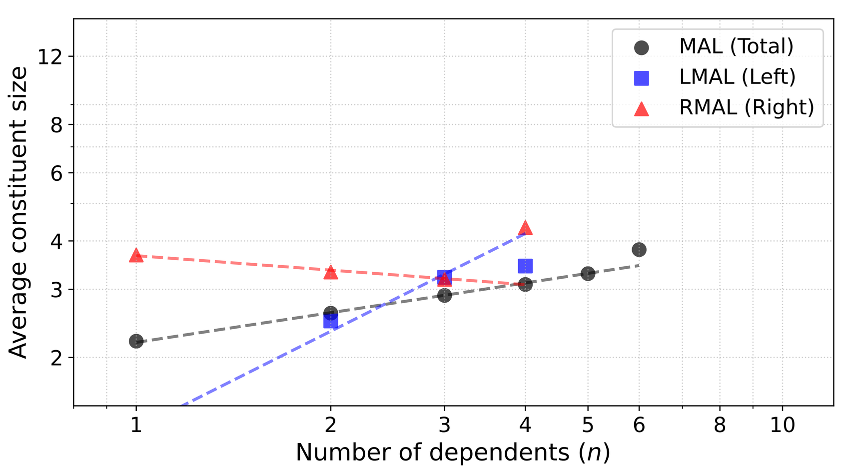
AS USUAL

German nails it. Naija wobbles. Arabic surprises. You?



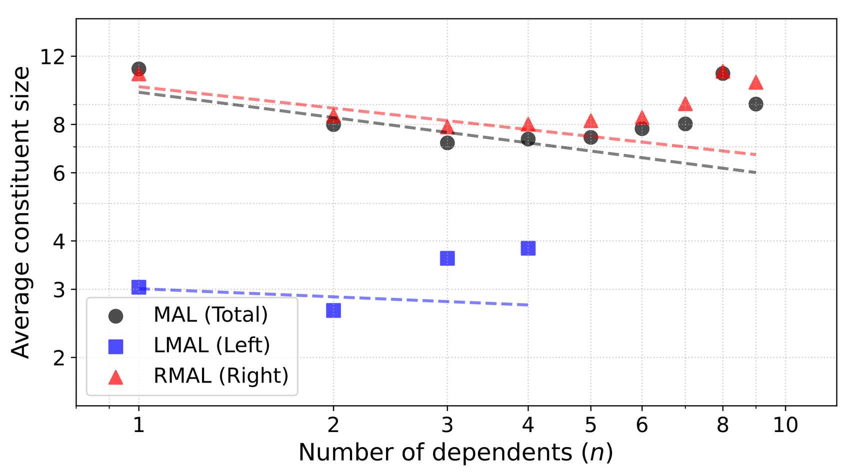
German – textbook MAL

slope = -0.17, R^2 - 0.99 on the log-log fit. A near-perfect power law: every extra dependent shrinks the average constituent. **The cleanest fit in the sample** – not the strongest slope, but the most orderly one. Is anyone surprised?



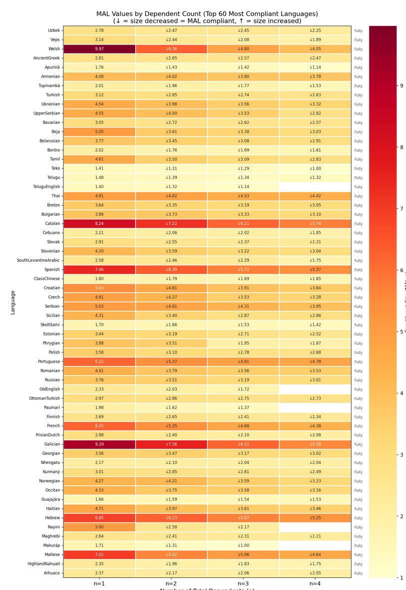
Naija – mixed, asymmetric

Left and right dependents diverge wildly. A young pidgincreole refusing to fit one curve – in fact the **strongest Anti-LMAL in the whole sample** ($\beta = +0.73$ preverbally).



Arabic – complex behavior

Arabic has a particularly low R^2. It follows MAL from $n = 1$ to 3, but becomes Anti-MAL from $n = 3$ onwards.



The wall of shame

Every row is a language, every column is the number of dependents n. **Color intensity** shows average constituent size (red is larger, blue is smaller). The arrows (↓, ↑) tell you exactly where the law holds or quietly fails.

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